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Realtime Collaborative Code Editor

Project Guide:
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Introduction

- Today, school's students have a very limited exposure to coding environment. Although they are taught the subjects like, python and java but still the tradational practices of writing on the paper are followed rather than practicing on a real-world environment.
- With the rapid growth of technology along with AI, the knowledge of technology is not only important in engineering fields but also in commerce and art fields also.
- Existing solutions to this problem either have a very complicated UI/UX which could be overwhelming for beginners or might not have a realtime collaborative coding environment.

Introduction (contd...)

- This project, a **Realtime Collaborative Code Editor**, provides a very simple solution to this problem where students can code, learn collaboratively in their browser itself without requiring any heavy installation.
- It provides a collaborative environment where students can code not only in class but also in holidays. It will help them to practice and work on their holiday homework projects.
- It is not only beneficial for students but also for teachers where they can explain the coding concepts to students collaboratively along with the workspace provided. This will also help them to monitor students progress in real time.

Key Features

- Simple and user-friendly UI/UX.
- Direct browser access without any heavy installation.
- No account creation required.
- AI chatbot for instant guidance in algorithms.
- Real time notifications including screen time notification for health.
- Communication via text/voice chat and also a shared drawing workspace for teachers.
- Real-time collaboration on coding projects using room creation.
- Ability to download files and also upload files locally.

Scope

Realtime Collaborative Coding

It uses WebSockets to provide a platform ere multiple students can create, practice and even run code directly in a browser window reachable through link, without any heavy installation.

AI-Assisted Learning

AI chatbot has been integrated which have the capability to solve any query of students and also guide them with hints and explanations, thus, making programming more interactive.

Scope (contd...)

Teacher–Student Collaboration

It will allow teachers to take coding classes online in a very interactive way and also provides workspace to teachers for concept explanation.

Holistic Learning Environment

It has the capability to go beyond coding by providing features like text/voice chat, screen-time management, and collaborative project-building which makes it complete ecosystem for coding education.

Learning Resources

Students can demand specific video lessons according to their needs which will be published on the platform regularly.

Objective

- **Provide Real-World Coding Exposure:** The main objective is to provide a coding environment to the students without any heavy installation so that they can practice in their school labs and even at home during holidays.
- **Promote Collaborative Learning:** It have features to enable collaborative coding thus helping students to work together on any project which in turn improves their sense of teamwork and problem solving skills.
- **Support Teachers:** It also provides teachers a drawing workspace to give them ability to explain concepts more perfectly to students.

Objective (contd...)

- **AI Assistance:** AI chatbot has been integrated to solve any query of students and also guide them with hints and explanations, thus, making programming more interactive
- **Healthy Digital Habits:** To notify students as soon as their screen time exceeds a limit of 2 hours to ensure a good and healthy use of technology.
- **Complete Learning Ecosystem:** This compherensive solution combines coding, collaboration, communication (text/voice), learning resources (like requested videos), health management and AI assistance into a single platform.

Literature Review

S.No	Author	Year	Methodology	Results
[1]	Aditya Kurniawan, et al.	2015	CodeR (Web-based editor using WebSocket, terminal integration)	Enabled collaborative coding with chat, terminal, and support for C, C++, Java
[2]	Shivam Chowdhury, et al.	2025	Real-time collaborative editor using WebSocket for interviews	Efficient platform for technical assessments with chat and code execution
[3]	Khushwant Virdi, et al.	2023	Review of collaborative editors; WebSocket + OT algorithms; prototype built	Identified major issues in latency, synchronization, and conflict resolution
[4]	Mohd Saad Ansari	2025	MERN stack + Socket.IO based minimalistic editor	Lightweight, simple, effective for peer coding and interviews

Literature Review

S.No	Author	Year	Methodology	Results
[5]	Anup Lal Yadav, et al.	2024	WebSocket integrated prototype for multi-user coding	Improved synchronization and collaboration features
[6]	Krishna Aher, Prof. Jagruti R. Mahajan	2023	React.js + Node.js + Socket.IO, with syntax highlighting, formatting, versioning	Enhanced productivity, proposed AI/ML integration for future
[7]	R. Pathak, et al.	2024	V-Code (Nuxt.js + Python Flask + OpenAl models)	Provided Al-powered suggestions & auto-completion for real-time coding

Research Gaps

Accessibility and Affordability

Many school students do not have access to personal laptops or devices and usually depend on outdated computer labs which create a gap in providing an affordable, lightweight, browser-based solution which can works seamlessly without heavy installations.

Real-World Exposure

Current school curriculums fail to provide students with real-time, hands-on exposure to coding and project making. Another gap exists in platforms that allow collaborative project work and coding practice during classes as well as holidays.

Research Gaps (contd...)

Teacher Involvement

While many coding platforms exist, but none are designed by keeping teachers in mind. There is a huge gap in tools that allow teachers to conduct coding classes and explain concepts interactively.

Privacy and Safety

Since the platform involves collaboration and communication (chat/voice), ensuring screen-time management of students is still a critical part which is overlooked by existing tools.

Methodology Used

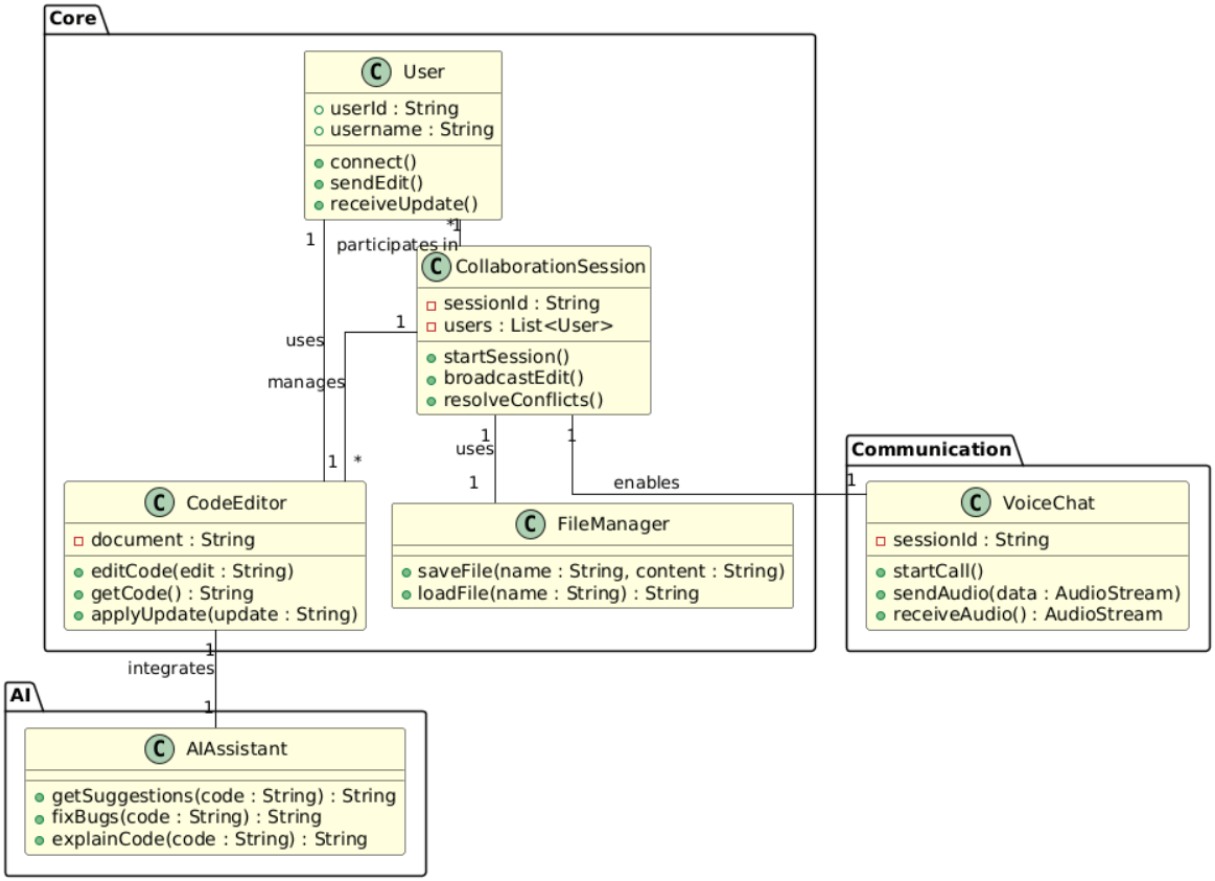


Figure: Class Diagram

Methodology Used

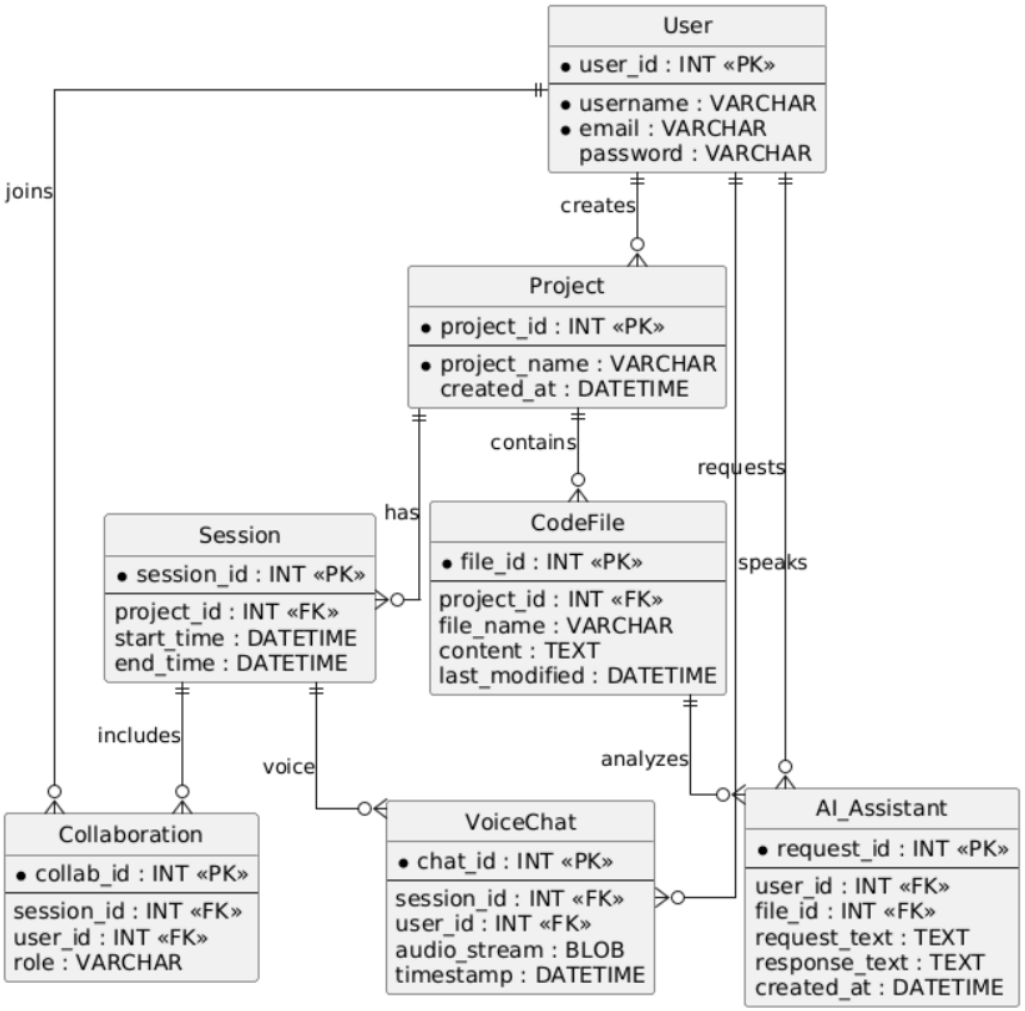


Figure: Entity-Relation Diagram.

Methodology Used

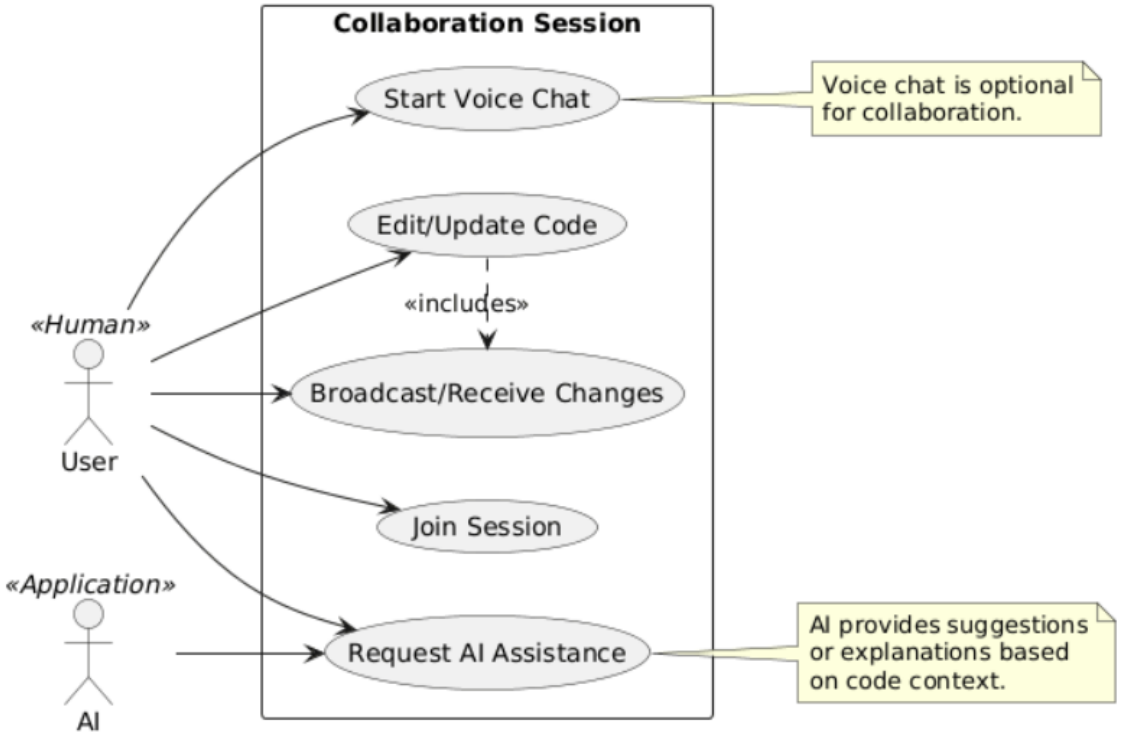
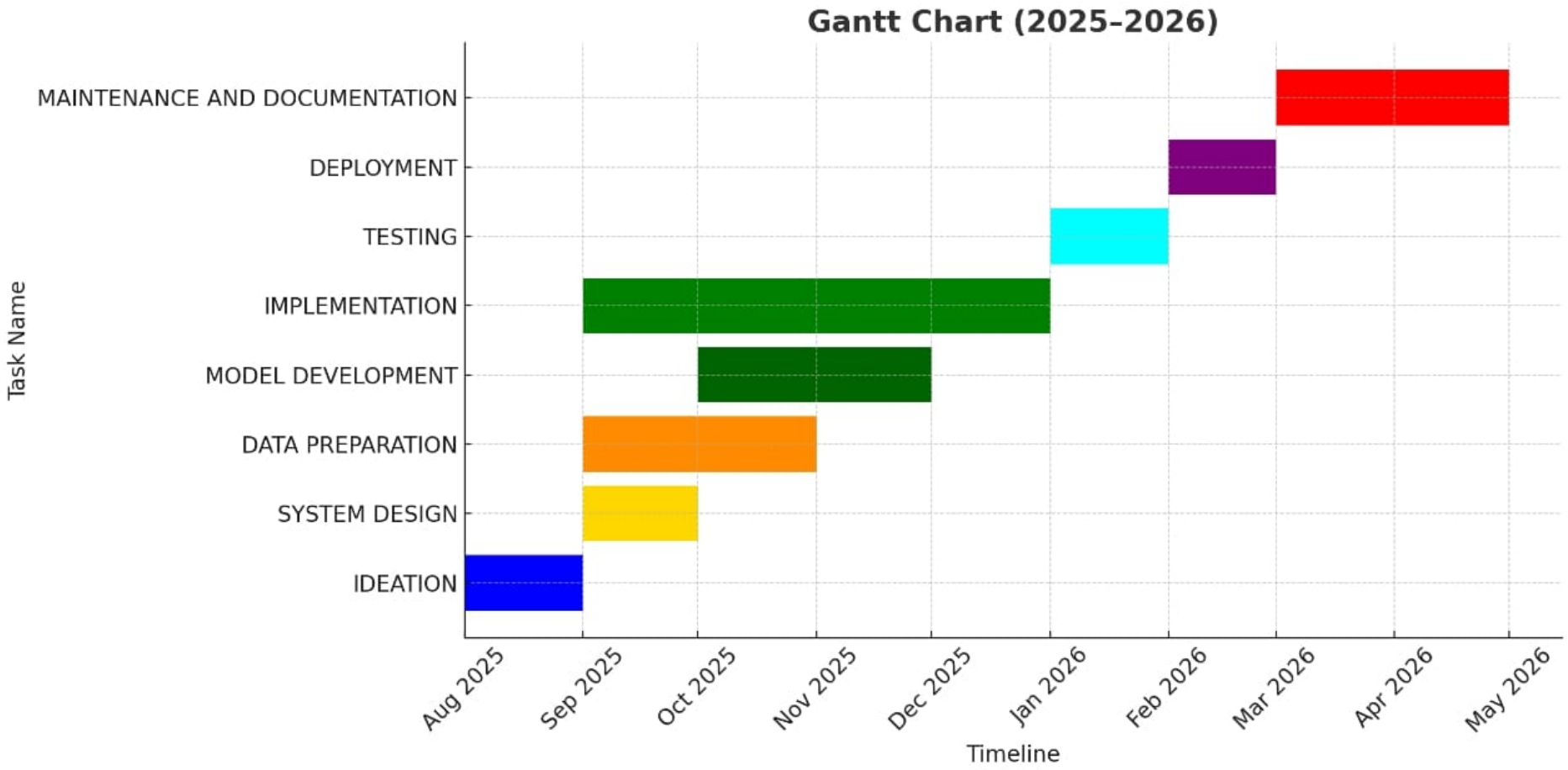


Figure: Use Case Diagram.

Gantt Chart


Gantt Chart (2025–2026)
MAINTENANCE AND DOCUMENTATION
DEPLOYMENT
TESTING
IMPLEMENTATION
MODEL DEVELOPMENT
DATA PREPARATION
SYSTEM DESIGN
IDEATION
Task Name
Aug 2025
Sep 2025
Oct 2025
Nov 2025
Dec 2025
Jan 2026
Feb 2026
Mar 2026
Apr 2026
May 2026
Timeline

Sustainable Development Goals Achieved by Project

SDG 4: Quality Education

The project enhances digital literacy by giving students real-time coding practice, AI chatbot support, and collaborative tools. It directly contributes to **Target 4.4**: increasing technical and vocational skills relevant for future jobs.

SDG 9: Industry, Innovation, and Infrastructure

By integrating AI and machine learning with education, the project fosters innovation and scalable infrastructure for digital learning. This supports **Target 9.5**: strengthening scientific research and advancing technology development.

Sustainable Development Goals Achieved by Project

SDG 10: Reduced Inequalities

The platform reduces education gaps by offering affordable, browser-based coding access to underserved regions with limited infrastructure. It addresses **Target 10.2**: empowering and promoting inclusion of all learners.

SDG 17: Partnerships for the Goals

The project builds collaboration between students, teachers, researchers, and institutions to promote shared learning and innovation. This fulfills **Target 17.6**: enhancing global partnerships for technology and knowledge exchange.

Result and Discussion

1. **Realtime Collaborative Code Editor**, has been designed to equip students with the practical exposure in real-time coding environment. It provide various features like collaborative coding sessions, AI-assisted guidance, and interactive features like voice/text chat and a shared drawing workspace for teachers. It will also give notifications to students if their screen time exceeds a limits of 2 hours.
2. The website's landing page has been made very simple and user-friendly for easy navigation. Right after joinng the room, students can see clear sections for coding projects, collaborative sessions, and AI assistance making it easy to start coding immediately without the need of any account.

Result and Discussions

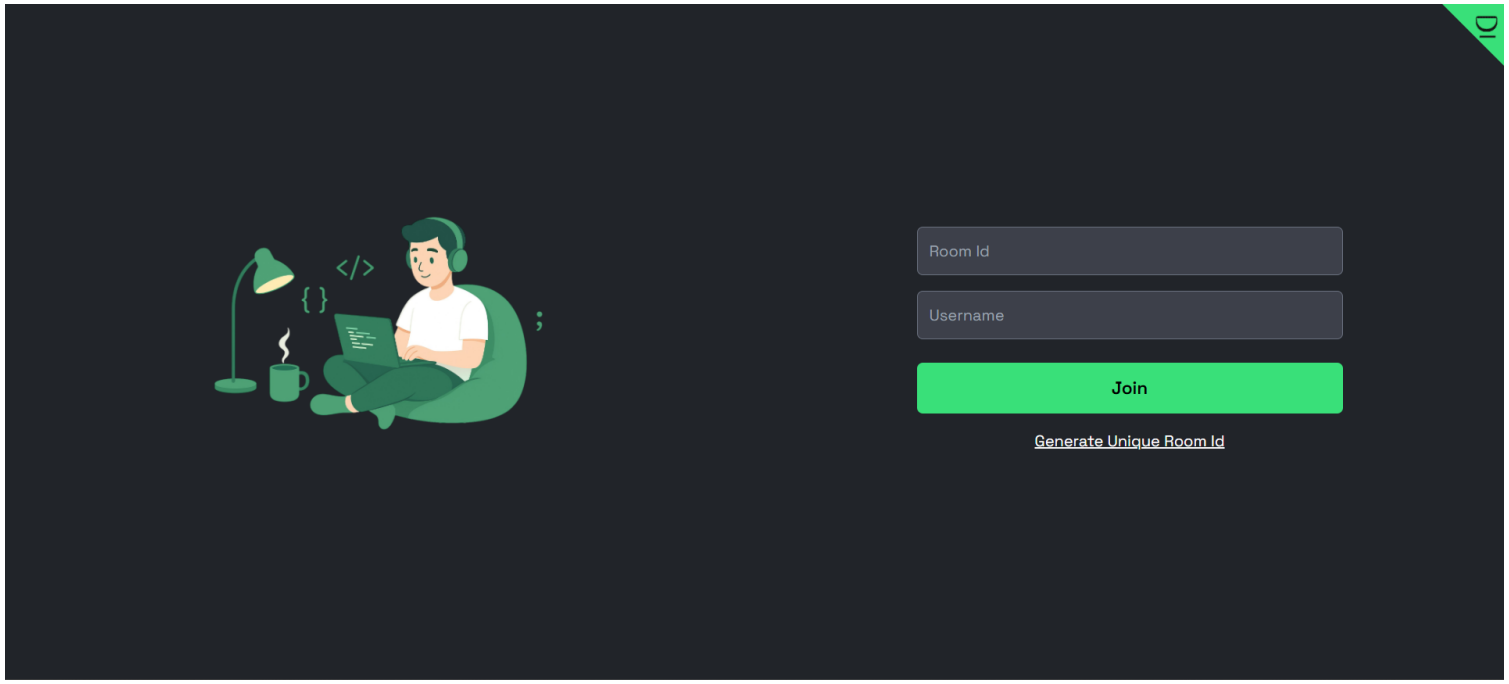


Figure: Home Page

Result and Discussions

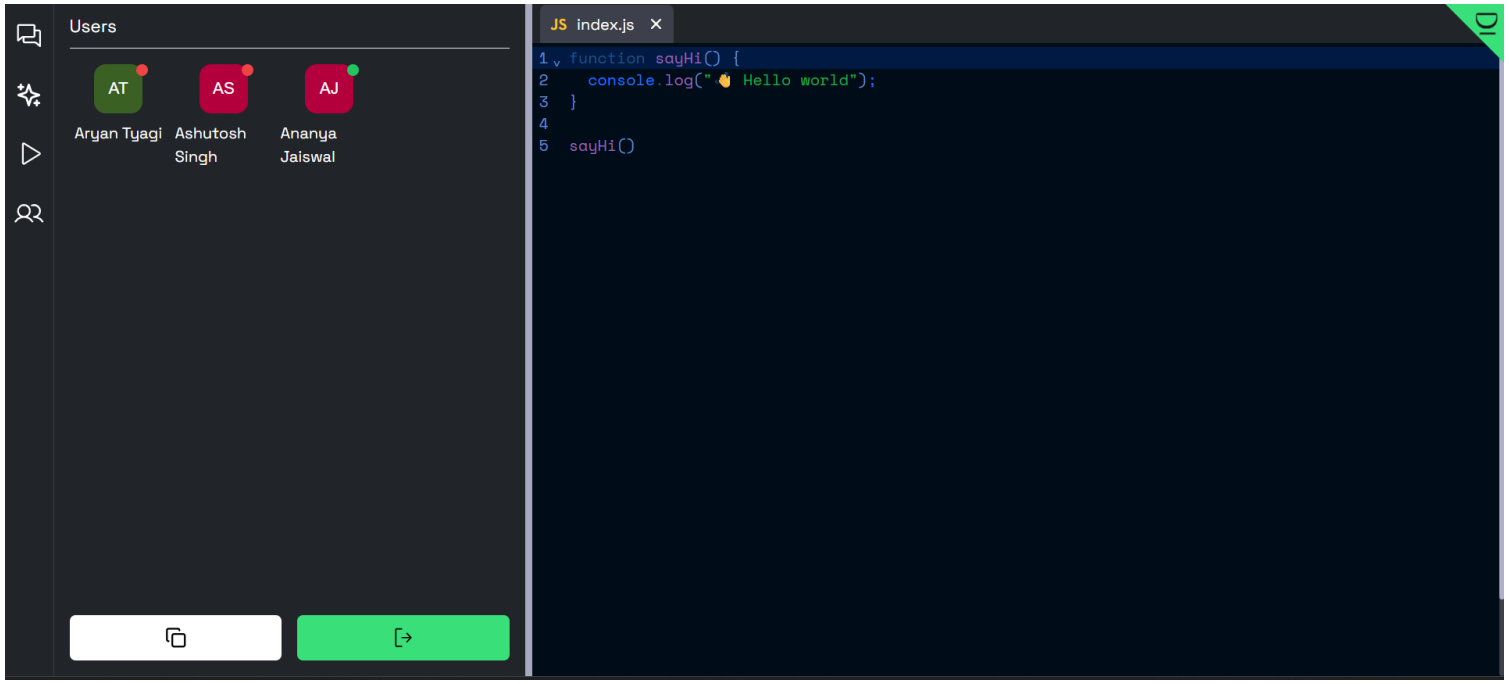


Figure: Home Page

Result and Discussion

1. The landing page has been designed for instant access to coding. It has a clean layout which highlights various collaborative tools, AI guidance, making navigation very simple and intuitive for learners of all ages.
2. Our moto "**Code & Chill**" do emphasizes a relaxed yet productive coding environment, where students can experiment, learn together, and get real-time feedback from peers and AI, which foster their creativity and teamwork.

Further Work

1. **Concept Development:** Define the core idea of "Code & Chill" by identifying key features like real-time collaboration, AI assistance, and student-friendly interface. Conduct surveys with students and teachers to understand their needs and refine the project vision.
2. **UI/UX Design:** Create intuitive wireframes, interactive mockups, and a clear navigation flow. Focus on simplicity so that students of all ages can easily access coding sessions, collaborative projects, and learning resources without confusion.

Further Work

1. **Backend Data Setup:** Design the server infrastructure to handle multiple real-time coding sessions, store project files securely, and manage user interactions such as chat and collaborative editing.
2. **AI Integration:** Incorporate the AI chatbot to provide instant coding guidance, suggest improvements, and answer students' queries. Continuously update AI suggestions based on user feedback.

Further Work

1. **Testing Feedback:** Conduct extensive testing of real-time collaboration, voice/text chat, AI responses, and coding functionality. Gather feedback from students and teachers to improve performance and usability.
2. **Enhancements:** Add features like video tutorials, problem sets for holidays, and project templates. Ensure notifications for screen time limits and achievements to encourage healthy coding habits.

THANKYOU